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Palaeomagnetism of coast-parallel alkaline dykes from western Norway; ages of magmatism and evidence for crustal uplift and collapse

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Summary. Palaeomagnetic investigation of 18 coast-parallel, alkaline dykes from the Sunnhordland region, western Norway, reveal the presence of multi-component magnetization in half of the dykes. The distribution of reversely magnetized directional end-points corresponds to a Kiaman (Permian) geomagnetic field direction, but the intersection of great circle trends coincide with the well-grouped and stable normal magnetization present in the remaining nine dykes. The stable remanence deviates significantly from any known palaeomagnetic field direction for western Europe and, although a non-dipole field/spot-reading origin cannot be ruled out, it is tentatively suggested that the magnetically stable dykes were intruded when the Sunnhordland area was affected by a temporary eastward tilt associated with a phase of crustal uplift over the adjacent North Sea area. A tilt-connection of 15° will bring the stable remanence directions in close agreement with the Triassic/Jurassic pole position for western Europe, implying a Kimmerian age of these intrusions. The range of reported K-Ar ages of some of the dykes is not in accordance with the palaeomagnetic results and is tentatively attributed to: (a) partial argon loss in the reversely magnetized dykes (Kiaman) caused by reheating during the intrusion of the assumed younger sequence of the magnetically stable and normally magnetized dykes, (b) excess argon in the assumed younger dykes, inherited from xenoliths present. A Kiaman intrusive age of all the dykes cannot be excluded, however, in which case a complete remagnetization of some of the dykes, occurring possibly during the proposed event of crustal uplift, was associated with partial loss of argon.

Introduction

In the Sunnhordland region, western Norway, a suite of more than 60 coast-parallel alkaline dykes intrude both the Precambrian basement and the overlying metamorphosed Lower Palaeozoic sedimentary and igneous rocks (*cf.* Fig. 1). K-Ar measurements on amphibole and whole rock samples indicate three episodes of dyke intrusion, occurring around 275, 220 and 160 Myr (Færseth, Macintyre & Naterstad 1976). These dykes represent the

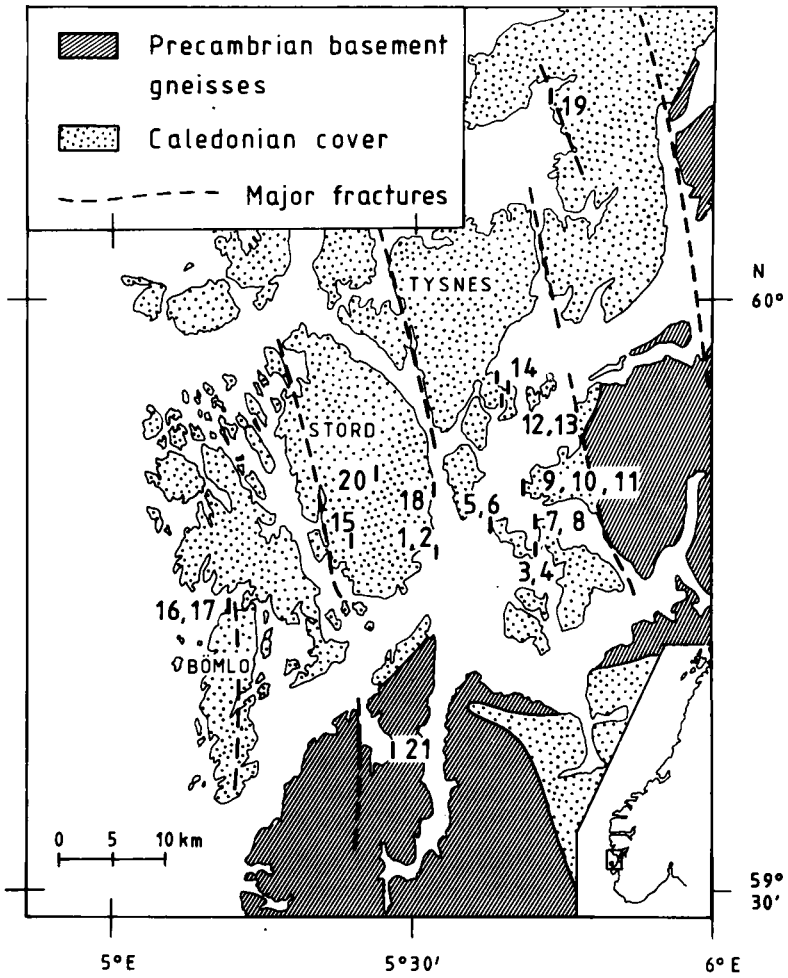


Figure 1. Simplified map of the Sunnhordland sampling region. Numbers refer to sampling sites.

youngest known igneous events in Norway and have been related to magmatic events associated with the development of the adjacent North Sea Region (Færseth *et al.* 1976).

As the Mesozoic palaeomagnetism for northern Europe is not very well established, oriented drillcores and hand samples were collected from 18 dykes (21 sites, *cf.* Fig. 1). Azimuthal orientation by both Sun compass and magnetic compass on approximately half of the samples (61) defines a mean magnetic deviation of $-6.4° \pm 0.9°$ the low scatter of which suggest only small variations of the geomagnetic field within the sampling area.

Magnetomineralogical properties

Polished sections from each site were examined in reflected light at magnification up to x 1900. The ore content ranges between 1–3 per cent by volume and is dominated by cubic to euhedral shaped, greyish-brown titanomagnetite grains, about half of which attract *magnetic colloid in a faint maze pattern. Grain sizes* range between submicron to larger than 100 μm. No exsolution lamellae of ilmenite were detected, free ilmenite appearing as thin, euhedral rods (1–10 μm x 10–100 μm) in about 30 per cent of the samples. With the exception of sites 16 and 17 which contain brown, euhedral titanomagnetite grains with no visible

alteration features, titanomagnetite grains show varying degrees of granulation features sometimes covering the grains completely. Bright, white intergrain bands of hematite (coating?) occur only rarely. Non-magnetic, euhedral pyrite grains, a fraction of which usually are significantly larger than the other opaques, are present in all samples.

Although the rocks give the visual impression of being rather fresh, the greyish colour and presence of granulation features on the titanomagnetite grains suggest that the dykes have been exposed to some degrees of low-temperature alteration (Ade-Hall, Palmer & Hubbard 1971), in accordance with the observation of partly serpentinized olivine (Færseth *et al.* 1976).

Saturation magnetization versus temperature, determined on at least two samples from each site, show characteristic features defining three different types of thermomagnetic curves (Fig. 2). The curves in Fig. 2(a, b) may be encountered within a single dyke, reflecting zones of different mineralogy (Hargraves & Petersen 1971). Reversible Curie points around 530°C are associated with irreversible curve shapes; the heating curves being more convex than the corresponding cooling curves. Hence, the initial magnetomineralogy has been subjected to some form of alteration upon heating in air to above 680°C, tentatively suggested to reflect the presence of cation deficient spinel (CDS) phases. The appearance of stable Curie points around 300°C upon cooling (*cf.* Fig. 2b) is attributed to the formation of a ferrimagnetic pyrrhotite-phase caused by the breakdown of pyrite.

The thermomagnetic curve in Fig. 2(c) is representative of one dyke only (sites 16 and 17). The single, initial Curie point around 420°C increases to around 480°C after heating associated with the appearance of a new magnetic phase with Curie point at 320°C. In basaltic rocks Curie points between 400–500°C very often reflect the presence of CDS-phases. On heating to above 600°C, however, CDS-phases are expected to unmix to a mineral assemblage incorporating almost pure magnetite associated with a corresponding increase in both Curie points (to 580°C) and saturation magnetization. However, the minor increase of the initial Curie point, to less than 500°C, in conjunction with the extrapolated decrease of the saturation magnetization of this phase and the creation of a low Curie point magnetic phase, suggest the presence of an initial relatively Ti-poor, unoxidized titanomagnetite, unmixing to a small proportion of a Ti-poor, titanomagnetite and a Ti-richer phase (Curie point ~ 320°C).

The acquisition curves of IRM in progressively increasing magnetic fields for natural and heated samples also reflect changes in the magnetic properties. While natural samples reach complete saturation IRM in fields of less than 0.2 T, associated with remanent coercivities between 9 and 19 mT, samples heated in air to above 600°C obtain only partial saturation in slightly larger fields. In addition, the intensities of $IRM_{0.2\,T}$ in untreated samples may increase by a factor of 2 after heating, suggesting the creation of magnetic phases with both higher coercivities and specific magnetizations. The magnetomineralogical properties do not

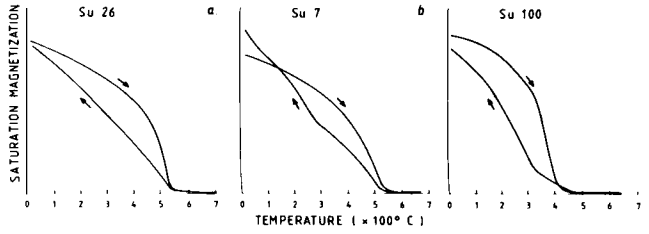


Figure 2. Characteristic thermomagnetic curves; (a) and (b) may both be encountered within a single dyke, (c) represent sites 16 and 17 only.

indicate any major differences between the dykes, with the exception of sites 16 and 17, and it is tentatively concluded that some form of cation deficient spinel is the dominant magnetic mineral phase.

Properties of remanent magnetization

Results of progressive thermal and alternating field demagnetization experiments reveal that half of the dykes (11 sites) carry a single, stable, normal (down) dipping distribution of remanent directions. The magnetization in the remaining nine dykes either show systematic directional trends, not necessarily along great circles, or an erratic non-systematic behaviour. Two distinctly different modes of directional behaviour are reflected by a narrow

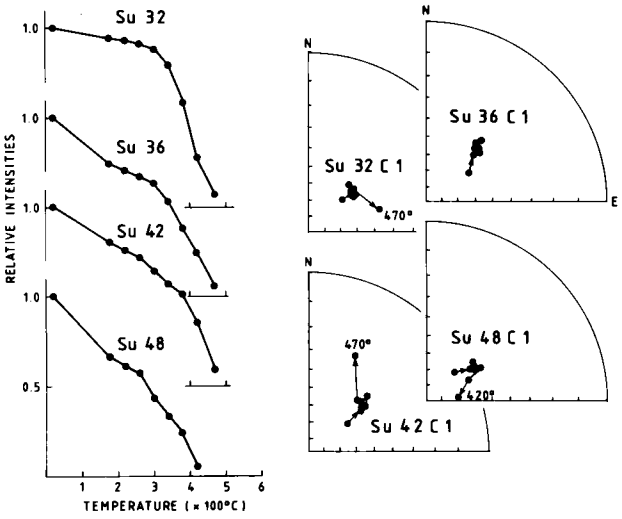


Figure 3. Characteristic thermal demagnetization curves and corresponding directional distributions for palaeomagnetically stable samples.

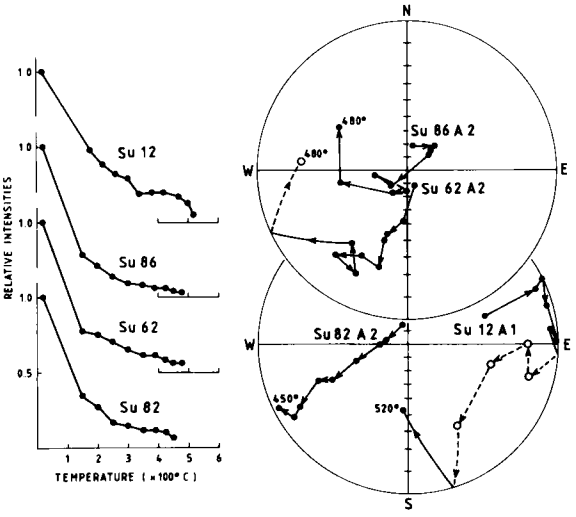


Figure 4. Examples of thermal demagnetization curves and corresponding directions for four unstable samples. Samples Su12 A1 and Su 86 A2 reflect non-systematic, erratic behaviour.

range of systematically lower values of the Q'_n-ratio (NRM/k) for the unstable rocks ($Q'_n \sim 0.4$), as opposed to a broad range of higher Q'_n-ratios for the stable rocks (mean: ~1.3). Also, the median destructive fields of the former are confined to a narrow range (around 4 mT) significantly lower than the MDF of the stable rocks (around 10 mT). Furthermore, upon thermal demagnetization the magnetization of the directionally stable rocks decays in a square shape to almost linear manner (*cf.* Fig. 3) indicating a major range of blocking temperatures between 400–470°C, as opposed to the magnetization of the unstable rocks which is reduced by more than 50 per cent during heating to below 200°C (*cf.* Fig. 4). The absence of initial low Curie point phases associated with the large drop in NRM suggest the presence of a thermally 'soft' magnetic component in the unstable rocks The directional trends (*cf.* Fig. 4, Su 62 A2, Su 82 A2) suggest the presence of two or more distinctly different magnetic components residing in thermomagnetically indistinguishable magnetic phases, indicating either a VRM, PTRM or TCRM origin of at least one of the components.

Palaeomagnetic directions

The stable remanent magnetization encountered in half of the dykes shows a rather tight, circular distribution around the dyke mean direction (Fig. 5). Exceptions are sites 16, 17 and 18 in which more elongated distributions can be seen. Radiometric ages (Færseth *et al.* 1976), dyke mean directions and relevant statistical parameters (Fisher 1952) are presented in Table 1. The mean direction of all dykes, D: 47.8°, I: 48.9°, corresponds to a palaeomagnetic pole position situated at 117° E, 46°N, which is considerably to the west of the known apparent polar wander curve for north-west Europe. The deviation from the latter is mainly one in longitude, i.e. declination. As six of the stable dykes were oriented by Sun compass, orientation errors can be ruled out. In the remaining dykes, samples showing erratic directional behaviour, exemplified by Su 12 A1, Su 86 A2 in Fig. 4, are excluded from further discussions. Systematic directional trends are all associated with down-dipping NRM-directions moving southwards in some cases revealing upward dipping components at the end

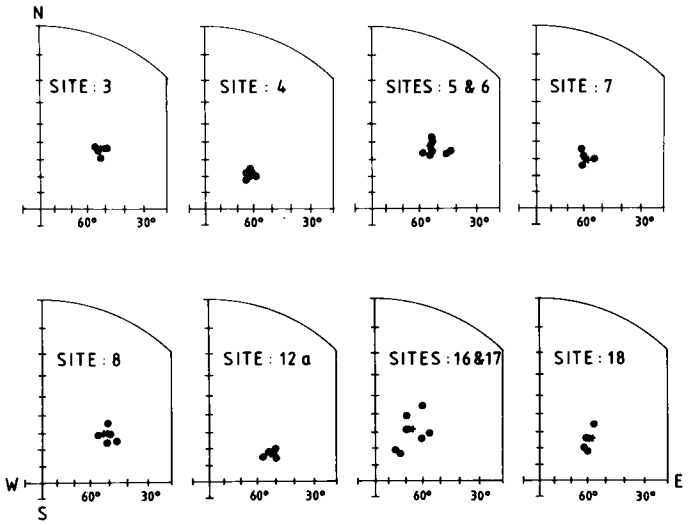


Figure 5. Stereographic plots of distributions of the stable remanent magnetization encountered in eight dykes. Dyke mean directions shown by asterisk. Site 12a represent a single thin dyke cutting through a wider, magnetically unstable dyke.

Table 1. Dyke mean palaeomagnetic directions after af/thermal demagnetization.

Site	K-Ar age* (Myr)	*N*	Declination	Inclination	κ	α_{95}
3	–	4	46.9	41.3	554.4	3.9
4	–	6	52.5	55.6	335.4	3.7
5,6	215	8	43.1	40.4	35.5	9.4
7	274	4	46.2	52.0	35.8	15.6
8	–	5	50.6	42.1	244.6	4.9
12	164	4	62.8	50.8	243.6	5.9
16,17	228	7	31.5	54.0	46.2	8.9
18	–	4	47.3	52.1	145.3	7.7
19	190	3	48.9	50.2	130.2	10.9
Mean direction of all dykes		(9)	47.8	48.9	106.4	5.0

*Færseth *et al.* (1976).
N: number of samples (dykes).
κ: precision parameter (Fisher 1952)
α_{95}: radii of 95 per cent circle of confidence (Fisher 1952).

of the demagnetization treatment. Only a few samples suggest stable up-dipping end points (*cf.* Fig. 6, Su17 A2, Su 21 A2, Su 56 A2). Great circles were successfully fitted to sections of 21 trend-patterns (Halls 1976), some of which are shown in Fig. 7 together with corresponding af-decay curves. Departure from a great circle fit at the highest demagnetization steps may reflect either the acquisition of stray components in the laboratory or the presence of more than two components being simultaneously demagnetized to different extents (*cf.* 62 A2 of Fig. 7 and Su10 A3 of Fig. 6).

Vector subtraction was attempted but was not successful in defining any stable distributions deviating significantly from the great circle intersecting point at *D*: 44°, *I*: 45°, which coincides remarkably well with the mean direction of the stable dykes, *D*: 48°, *I*: 49°. Hence, it is tentatively concluded that the pronounced great circle trends reflect the

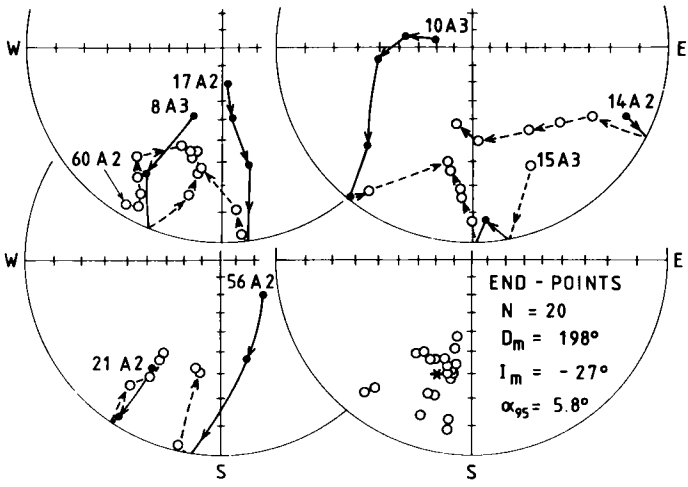


Figure 6. Examples of systematic directional trends present in unstable samples. Distribution of all up-dipping end-points shown in lower right stereogram.

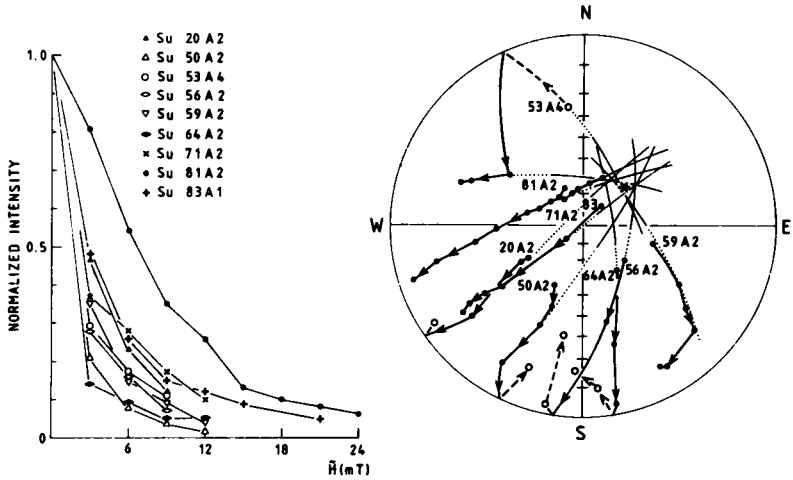


Figure 7. Characteristic examples of systematic directional trends along great circles and corresponding af-decay. Great circles intersection point shown by asterisk.

presence of two major components; one down-dipping normal component, essentially identical to the stable directions, and one shallower dipping component of reverse polarity. The inferred reversed component cannot be uniquely defined due to failure in reaching indisputably stable end points. However, the mean of all up-dipping, 'thermal' directions define an axis of magnetization that is significantly different from that defined by the normally directed remanence. The mean of all up-dipping 'end'-points (D: 198°, I: −27°) may be regarded as an estimate of the reversed component and corresponds to a palaeomagnetic pole situated at 43°N, 162°E, in excellent agreement with the established Permian pole of north-west Europe (see, e.g. Phillips & Forsyth 1972).

Discussion

The investigated dykes, carrying two distinctly different palaeomagnetic directions of opposite polarity, may represent two major intrusive episodes. The stable, down-dipping directions (pole A′, Fig. 8) is probably the youngest (and of a primary origin) overprinting an older, more shallower dipping reversed magnetization (pole B, Fig. 8) residing in the magnetically unstable dykes. While the latter magnetization corresponds to a pole coinciding with the Permian palaeomagnetic field for north-west Europe, the former, normal component, does not fit in with any known European palaeopole. Neither a spot reading origin nor a non-dipole field origin of the normal magnetization can be ruled out, but both interpretations imply the almost simultaneous intrusions of at least nine alkaline dykes, some of which show significant differences in the geochemical compositions (Færseth *et al.* 1976). The deep-seated magma sources, reflected by the occurrence of ultramafic inclusions and xenoliths, is also tentatively regarded as evidence against simultaneous intrusion of the nine dykes. It is therefore suggested that the normal magnetization represents a substantial time coverage of the geomagnetic field, and may to a sufficient approximation represent a true palaeomagnetic pole position.

The application of any tectonic correction which may bring the deviating pole into closer agreement with known palaeomagnetism for western Europe is highly conjectural in that the (pre-intrusive) Caledonian orogeny induced extensive tectonic and metamorphic alteration

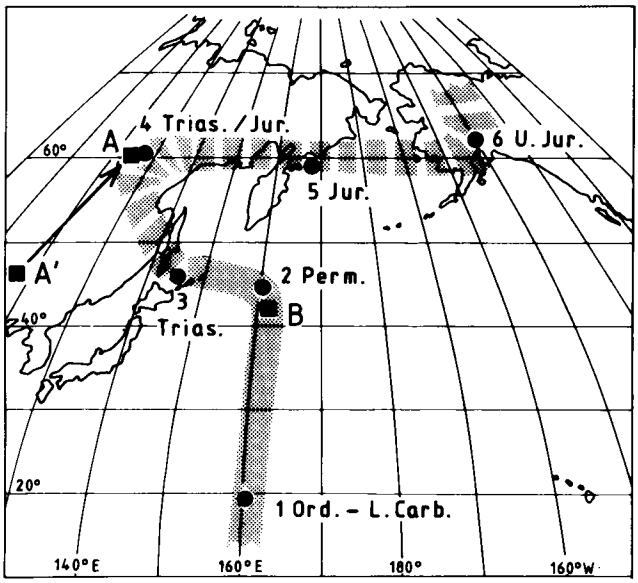


Figure 8. The Sunnhordland poles A' (A tilt-corrected) and B in conjunction with suggested polar wander curve for Europe. Poles 1–6 from following sources: 1 = Storetvedt (1968), 2, 3 = McElhinny (1973), 4 = Storetvedt (1978), 5 = Phillips & Forsyth (1972), 6 = Briseid & Halvorsen (1974).

in the country rock. No suitable geological feature is present which enables a unique, independent determination of any post-intrusive tectonic correction. The close agreement between the assumed older, reversed magnetization with the Permian field, however, suggest that the area has been exposed to some temporary tectonic movements, post-dating the oldest intrusive episode. The inferred Permian age of the latter suggest a connection with the evolution of the Oslo Graben fracturing and magmatism (Ramberg & Spjeldnæs 1978), which has been interpreted to reflect an aborted rifting event, incorporating plume-generated crustal uplifts initiated in Late Carboniferous and Early Permian times (Whiteman *et al.* 1975). It is likely that tensional stresses in the lithosphere, associated with crustal uplift, also affected the Sunnhordland region some 450 km to the west, causing the emplacement of the oldest (Permian) alkaline dykes more or less contemporaneously with the extensive, pre-Mesozoic volcanic activity in the Oslo region (Storetvedt *et al.* 1978). After a period of magmatic tectonic quiescence, it is proposed that magmatism in the Sunnhordland area was rejuvenated in conjunction with the Mesozoic alkaline rift volcanism in the North Sea (Howitt, Aston & Jacque 1975), reflecting a phase of crustal uplift over the present day Viking Graben and preceding an inferred crustal plate separation (Pegrum, Rees & Naylor 1975). During the Kimmerian crustal uplift, it is proposed that the Sunnhordland region was exposed to eastward tilting and tensional stresses in conjunction with which the youngest dykes were intruded. When rifting was aborted, reflecting the Upper Cretaceous commencement of the successful separation of the European and north American plates (Pegrum *et al.* 1975), subsequently post-intrusive block-faulting (Færseth *et al.* 1976) and crustal collapse brought Sunnhordland tectonically back to its present more or less pre-tilt position.

Dominating north–south striking fracture patterns, mapped from Landsat images (Gabrielsen & Ramberg 1979) cover major areas of coastal western Norway and are suggested to represent tensional features associated partly with the inferred crustal uplift

phase. Accepting a north–south strike for the eastward tilt, correcting for a dip of 15° to the east will bring the stable, down-dipping palaeomagnetic direction (pole A, *cf.* Fig. 8) in close agreement with the somewhat uncertain Lower Mesozoic section of the apparent polar wander curve for western Europe, represented by the inferred Late Triassic to Middle Jurassic pole from the Duncansby neck of north-east Scotland (Storetvedt *et al.* 1978, pole 4, *cf.* Fig. 8).

Although the suggested magmatic/tectonic history of the Sunnhordland region is compatible with available geophysical and geological data, there are significant discrepancies between the radiometric ages (Færseth *et al.* 1976) and the corresponding palaeomagnetic results. From Table 1 it is evident that the five radiometrically dated dykes (whole rock K-Ar data), carrying stable, coinciding palaeomagnetic directions of inferred Late Triassic/Middle Jurassic age, have K-Ar ages ranging between 164–274 Myr. During this time span, however, several reversals of the geomagnetic field occurred in addition to a significant apparent polar wandering. As the dykes in question do not show any reliable indications of the presence of multi-component magnetizations, they are concluded to carry a primary magnetization representing a unique direction of the palaeomagnetic field. In addition, five of the supposed older reversely magnetized dykes (Permian), having major directional overprints probably induced during reheating associated with the intrusion of the Triassic/Jurassic dykes, reveal K-Ar ages ranging between 214–254 Myr. The Kiaman reversed magnetic interval which terminated in Upper Permian (around 230 Myr) was succeeded by a mixed interval lasting some 25 Myr (McElhinny 1973). The close agreement between the estimated mean direction of the reversed end-points with the Permian field for western Europe strongly suggests that a majority of the magnetically unstable dykes were intruded within the Kiaman interval. It is tentatively suggested that some of the erratic directions may reflect the presence of mixed polarity components (Upper Permian to Lower Triassic?) which the applied techniques have not been able to resolve.

The serious discrepancies between the palaeomagnetic results and the ages determined by conventional K-Ar techniques apparently put some severe restrictions on the reliability of either of the two methods for dating purposes. However, the coinciding stable, single component magnetization residing in one suite of the dykes is not compatible with an acquisition time as suggested from the K-Ar technique. Hence, it is tentatively concluded that the applied radiometric method has not been adequate in determining the ages of dyke intrusion. As the K-Ar ages (Table 1) generally appear to be too old, this may be due to the presence of excess argon, either derived from the country rock or as inherited argon derived from the reported xenoliths present (Færseth *et al.* 1976).

An alternative interpretation of the radiometric age spread is that these dykes are of Kiaman age, but reflect incomplete retention of argon during a Mesozoic thermal event which on the other hand led to a complete remagnetization of the dykes in this group. At the present it is not possible to choose between these two alternatives.

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